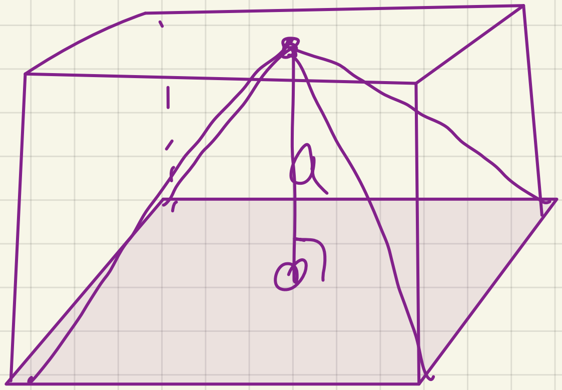
4



$$V = \frac{1}{3} S_{осн} \cdot h$$

$$V = \frac{1}{3} \cdot a \cdot a^2 =$$

$$= \frac{a^3}{3} = \frac{3^3}{3} = \underline{9}$$

6

$a \to a+1$

$S \to \boxed{S+54}$

$S = 6a^2$

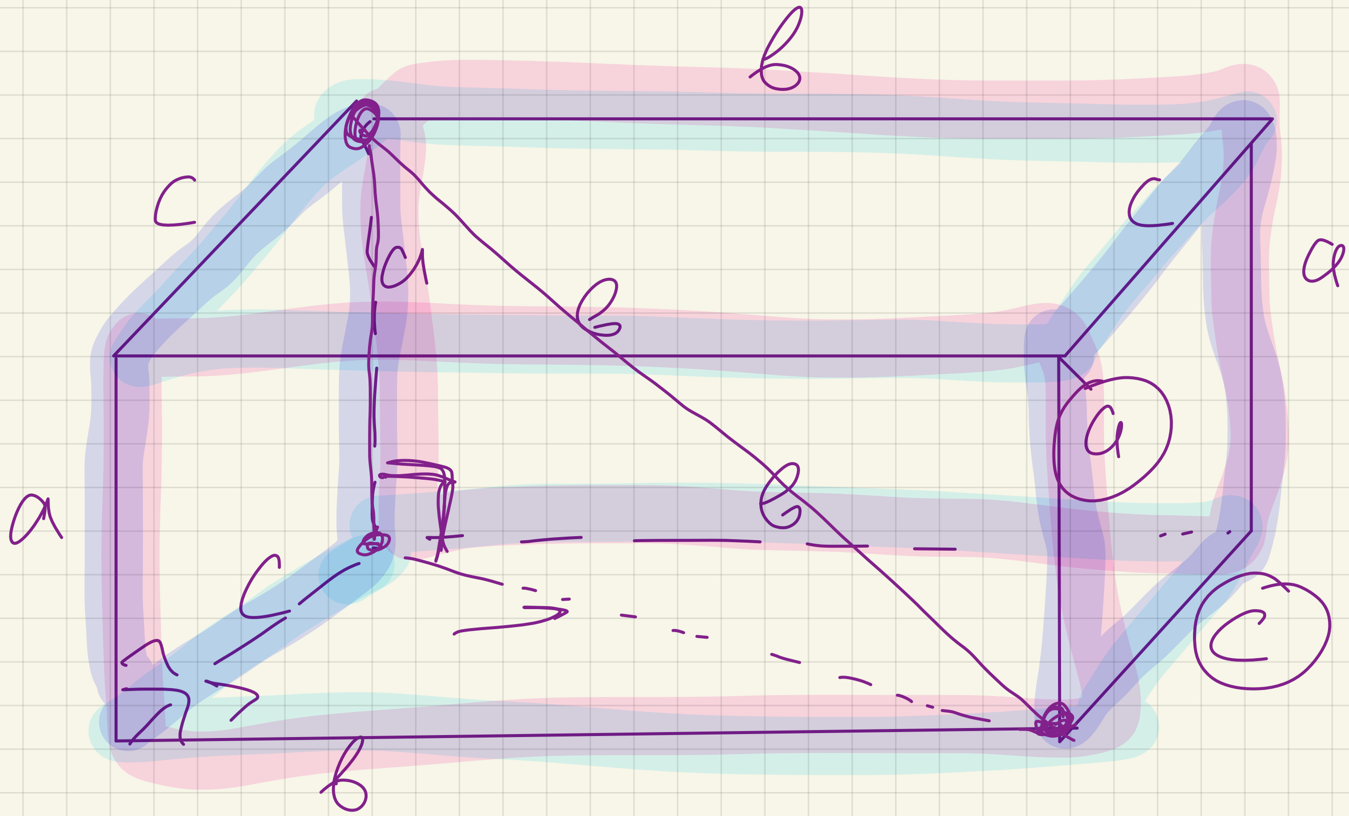
$S_1 = 6(a+1)^2 = S+54 = 6a^2+54$

$6(a^2+2a+1) = 6a^2+54$

$\cancel{6a^2} + 12a + 6 = \cancel{6a^2} + 54$

$12a = 48$

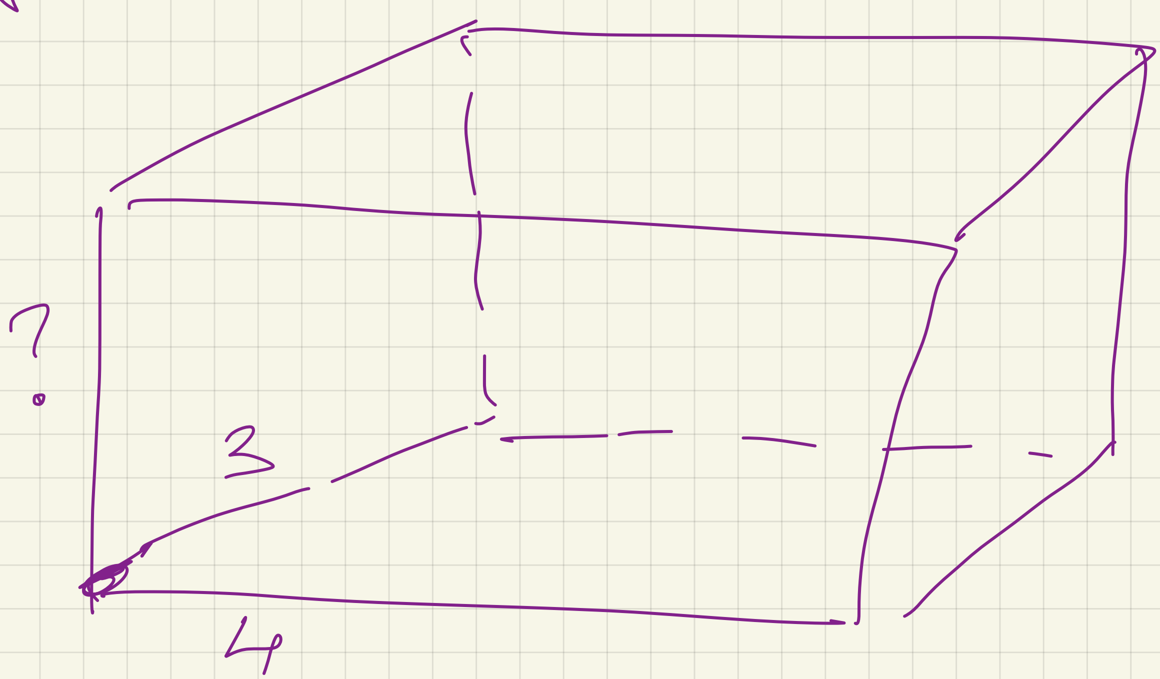
$a = 4$



$$V = a \cdot b \cdot c$$

$$S = 2ac + 2bc + 2ab = 2(ac + bc + ab)$$

$$\sqrt{a^2 + b^2 + c^2} = d$$



$S = 94$

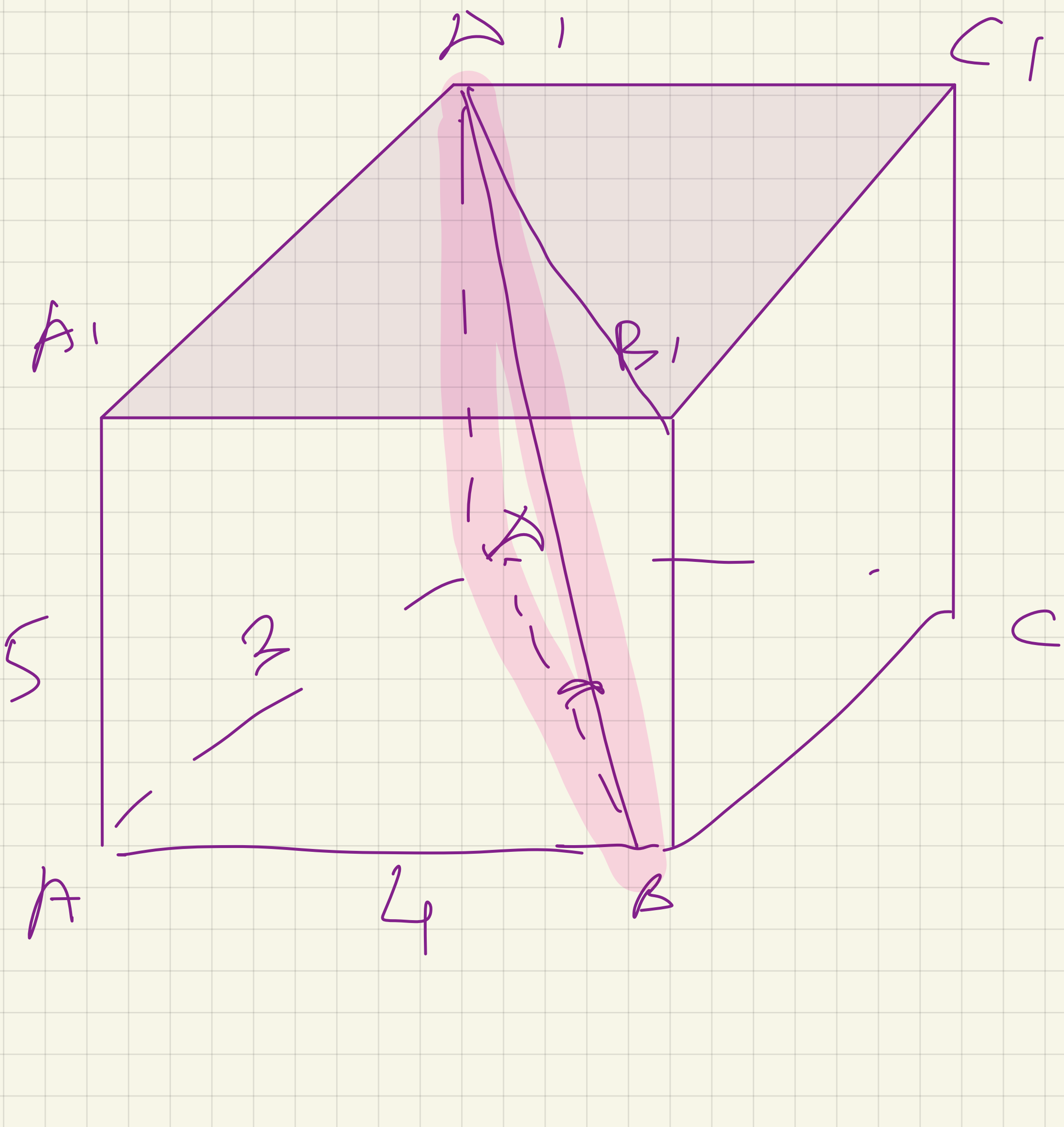
$2(3\cdot 4 + b\cdot 4 + 3\cdot b) = 94$

$12 + 7b = 47$

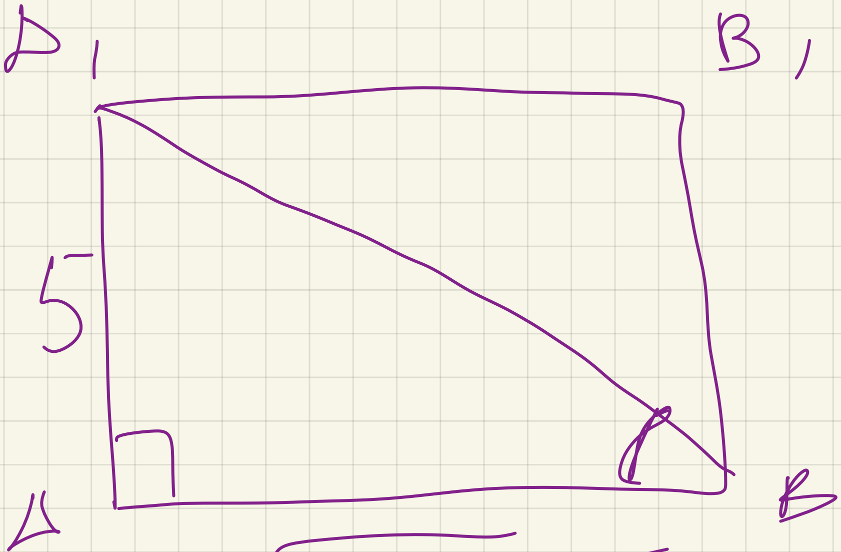
$7b = 47 - 12$

$7b = 35$

$b = \frac{35}{7} = 5$



D_1BD?



$45°$

$\sqrt{3^2+4^2}=5$

$AD = 2$

$AB = 4$

$d = 6$

$\rightarrow V?$

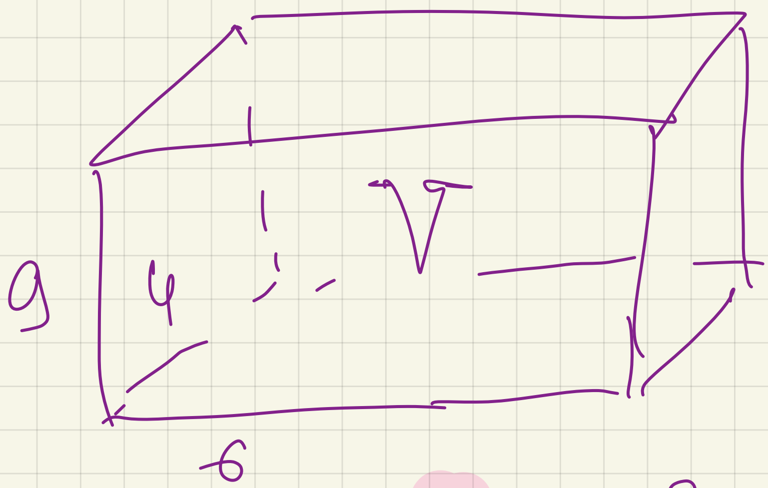
$2^2+4^2+x^2=6^2$

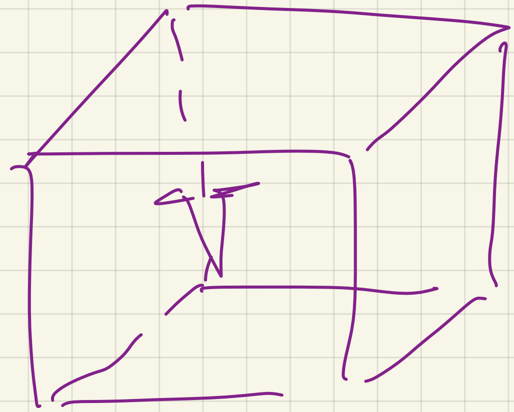
$4+16+x^2=36$

$x^2=16$

$x=4$

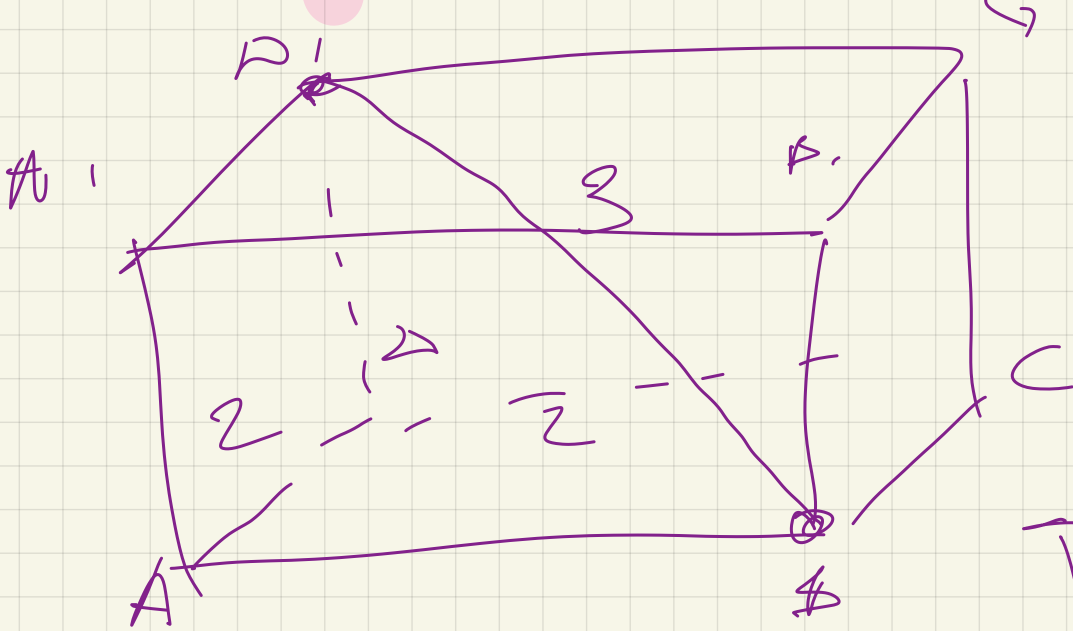
$V = 32$



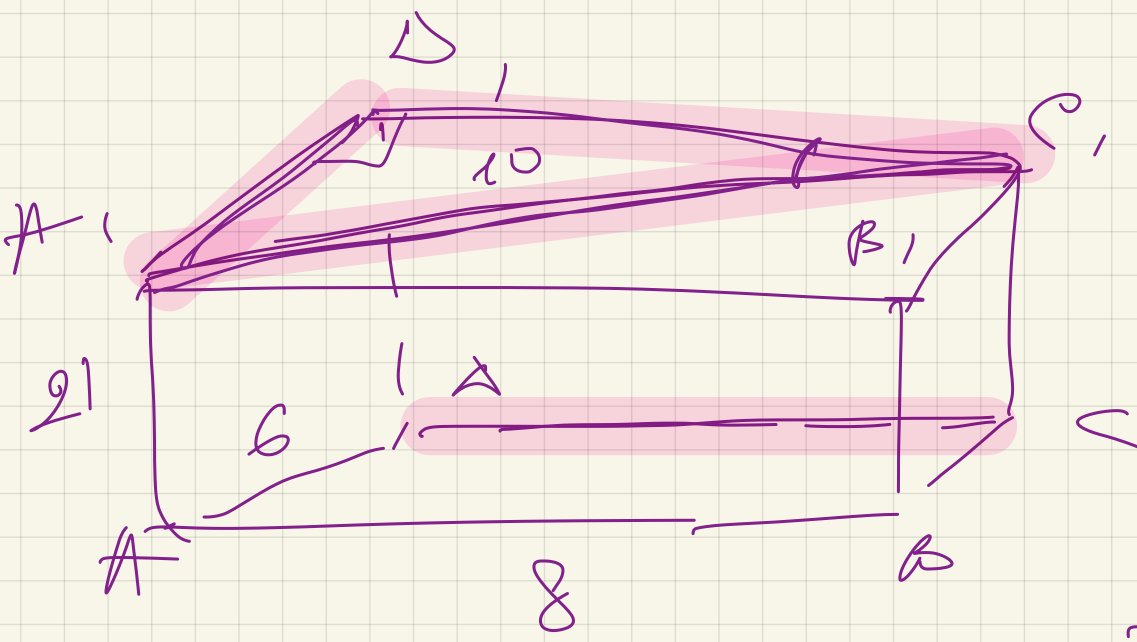


a?

$216 = a^3$

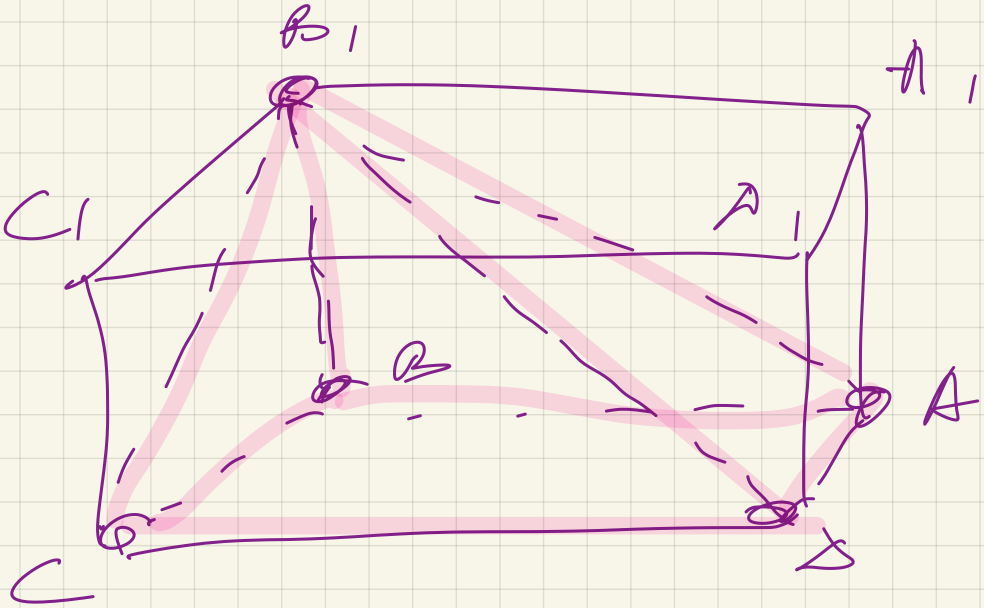


$V = $?



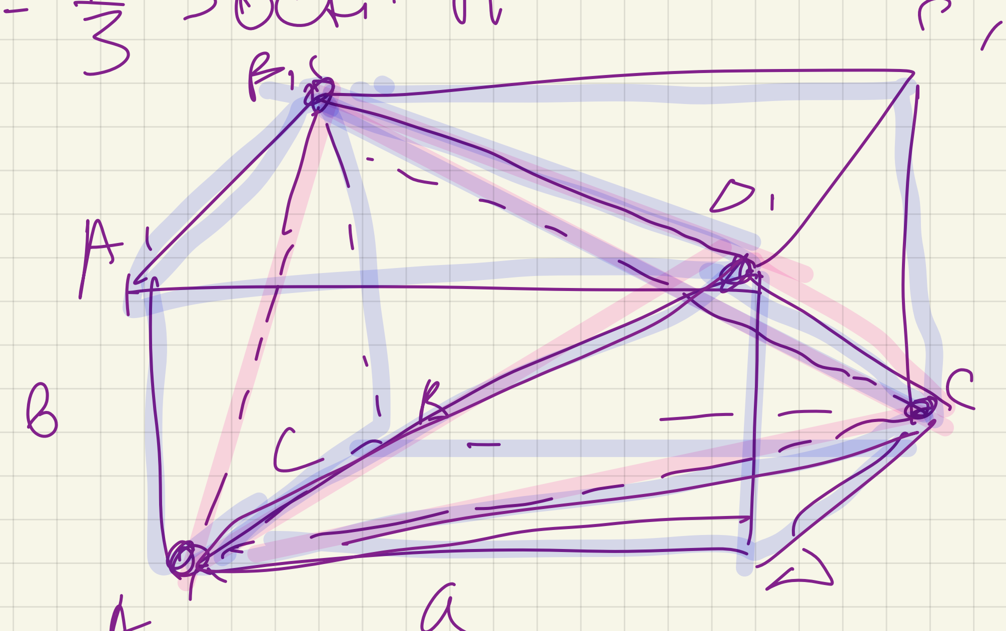
пресеч CD и A_1C_1 Sпр ?

$$\sin A_1C_1D_1 = \frac{A_1D_1}{A_1C_1} = \frac{6}{10} = 0,6$$



$AB = 9$
$BC = 5$
$BB_1 = 8$

$$V = \frac{1}{3} S_{осн} \cdot h$$



$V = 4,5$

$$V_{A_1B_1D_1A} = \frac{1}{3}\left(\frac{a \cdot c}{2}\right) \cdot b =$$

$$\frac{1}{3}\frac{a\cdot c\cdot b}{2} = \frac{1}{6}abc$$

$$V_{AD_1CD} = \frac{1}{3}\left(\frac{a\cdot c}{2}\right)\cdot b = \frac{1}{6}abc$$

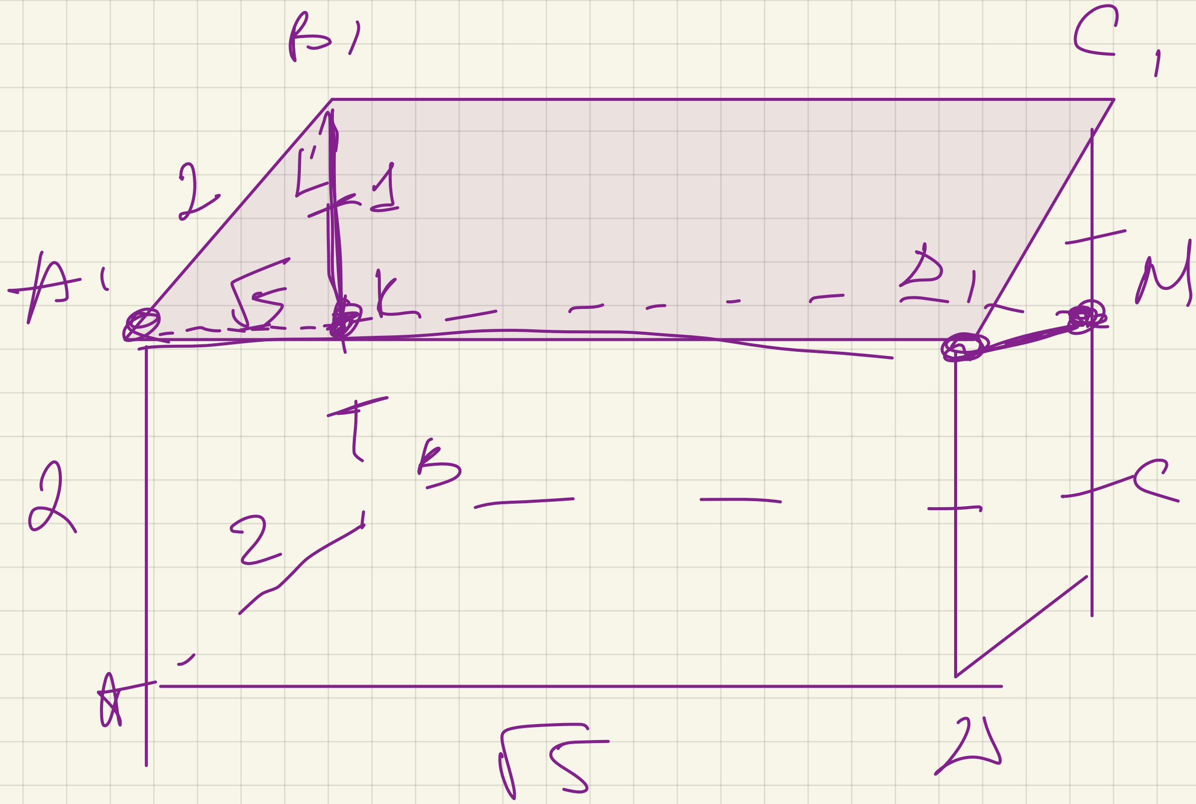
$$V_{A_1C_1CB_1} = \frac{1}{3}\left(\frac{ac}{2}\right)\cdot b = \frac{1}{6}abc$$

$$V_{ABCB_1} = \frac{1}{3}\left(\frac{ac}{2}\right)\cdot b = \frac{1}{6}abc$$

$$V = V_1 - \frac{4abc}{6} = abc - \frac{4abc}{6}$$

$$\frac{2}{6}abc = \frac{1}{3}abc =$$

$$\frac{4,5}{3} = 1,5$$



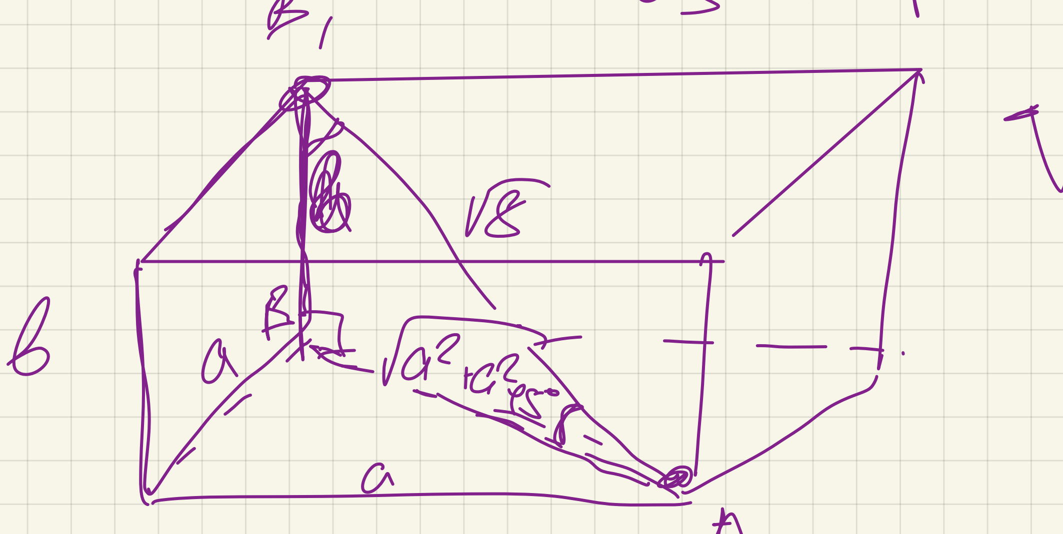
$$S_{A_1D_1K} = \sqrt{5}\cdot\sqrt{5} = 5$$

$$A_1K = ?$$

$$A_1D_1 = \sqrt{5}$$

одна грань — квадрат!

V ?



$$a^2+a^2+b^2=8$$

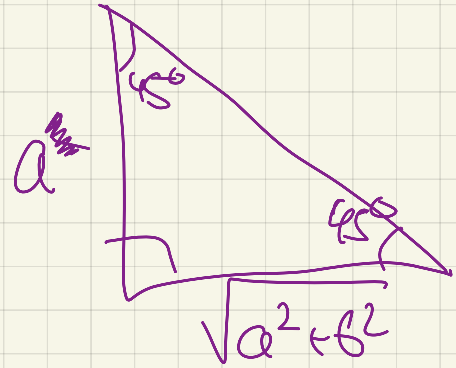
$$b=\sqrt{a^2+a^2}$$

$$b^2=\boxed{2a^2}$$

$$2a^2+2a^2=8$$

$$4a^2=8$$

$$a^2=2$$



$$a=\sqrt{2}$$

$$b=\sqrt{2a^2}=\sqrt{2\cdot 2}=2$$

$$a=\sqrt{2}$$

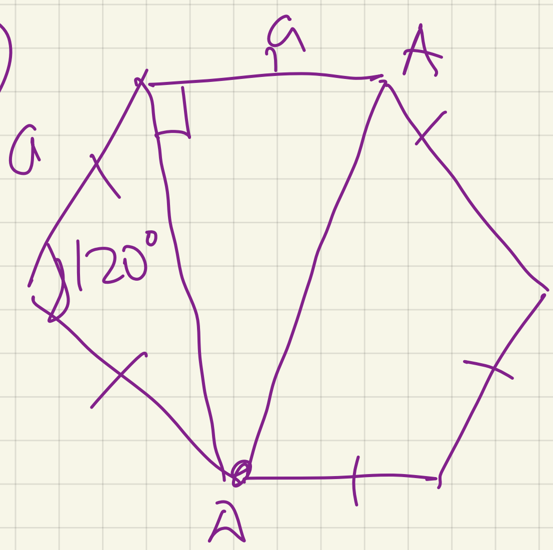
$$b=2$$

$$V=a\cdot a\cdot b=\sqrt{2}\cdot\sqrt{2}\cdot 2=4$$

$$a=\sqrt{a^2+b^2}$$

$$b=0$$

Д/З



S ?

AD ?